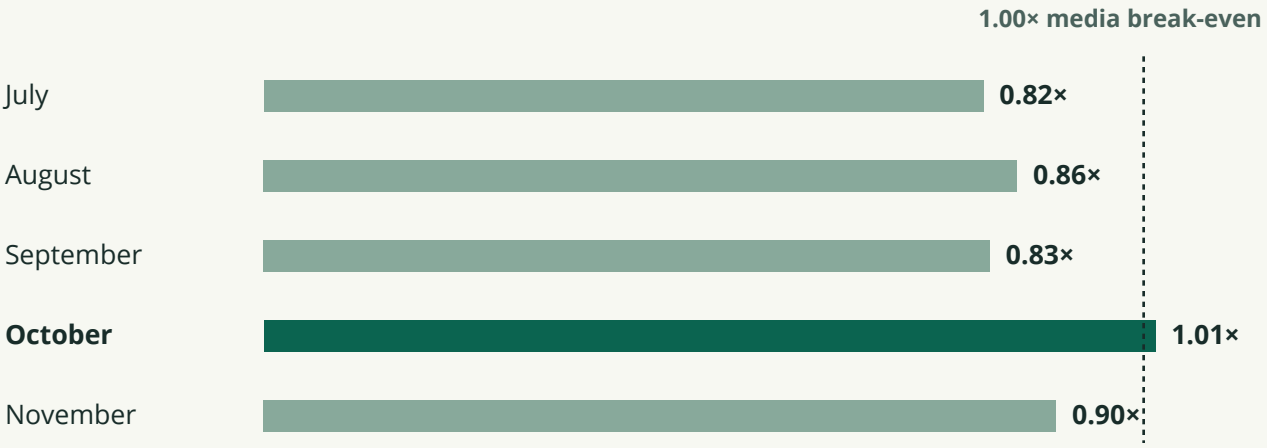
MARKETING CAMPAIGN PERFORMANCE & BUDGET OPTIMIZATION

Where should the next ad dollar go?

I used Excel and Tableau to evaluate five months of paid search and design a $10,000 test budget. November drove the most conversions, while October was the only month to cover its ad spend.

$635,372	$561,960	0.88×
Ad spend	**Tracked revenue**	**Return on ad spend**

Monthly return on ad spend



VOLUME AND RETURN

November recorded 43,267 conversions and 0.90× ROAS. Across all five months, tracked revenue fell $73,412 short of ad spend, before product and operating costs.

BUDGET ALLOCATION

What a $10,000 test buys

Promo Code led the recurring themes at 0.93× ROAS. I would use it as the lead test, reduce spend on Sale, and retest Black Friday/Cyber Monday separately. The seasonal result of 4.36× came from just $47 across two records.

Allocation	Budget	Historical ROAS	Modeled revenue
Promo Code	$4,000	0.93×	$3,730.77
Promo	$3,000	0.92×	$2,756.85
Coupon Code	$2,000	0.89×	$1,779.41
Reserve	$1,000	—	—

The deployed budget still falls short

At unrounded historical ROAS, $9,000 deployed would bring in $8,267.03 at 0.92× ROAS. Covering media cost would require $732.97 more revenue, an 8.87% increase over the modeled amount.

TEST BEFORE SCALING

Test offer wording or landing-page changes on Promo Code. Keep the $1,000 reserve for follow-up testing and compare incremental revenue and contribution margin before scaling.

KPI VALIDATION & LIMITATIONS

How I checked the return

Rebuild the metrics from counts

The source conversion-rate column did not consistently match conversions divided by clicks. I used raw counts and amounts for CTR, conversion rate, CPC, CPA, ROAS and media ROI. Rollups divide sums rather than averaging row-level ratios.

Check what spend is associated with

Across the 190 ad-group-month observations, spend correlated strongly with tracked revenue ($r = 0.99$) and had almost no linear relationship with row-level ROAS ($r = -0.04$). These descriptive correlations cannot estimate the effect of incremental spend.

Define break-even clearly

Media ROI is (tracked revenue minus ad spend) divided by ad spend. A 1.00× ROAS covers media cost alone. Product margins, operating costs, customer lifetime value and incrementality are unavailable.

Treat the allocation as a scenario

The Excel scenario applies each promotion's historical ROAS to the proposed allocation. It illustrates how much improvement the mix would still need. A controlled test with predefined outcomes would be needed before changing the ongoing budget.

DATA SOURCE

Shopping Mall Paid Search Campaign Dataset · Kaggle